

SAMARTH SACHAN

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EDUCATION

Year	Degree / Exam	Institute	Grade
2019 - 2024	BT-MT, Mechanical Engineering	Indian Institute of Technology Kanpur	BT: 8.3 MT: 9.5
2018 - 2019	XII (ISC)	Dr. Virendra Swarup Education Center, Kanpur	95.50%
2016 - 2017	X (ICSE)	Dr. Virendra Swarup Education Center, Kanpur	95.00%

WORK EXPERIENCE

Control Engineer | Automated Motion Group | Jaguar Land Rover Jul 2024 – Present

Jaguar Land Rover - Technology Business Services India | Bangalore, India

- Contributed in development and testing of **Terrain Response** (TR), **Auto Terrain Response** (AutoTR), **Configurable Terrain Response** (CTR), and **Intelligent Driveline Dynamics** (IDD) applications, working across the **ASPICE Software V-cycle**
- Authored **Software Requirements** and linked them to Simulink models to improve software **traceability** and requirement coverage
- Developed Service-Oriented Simulink models following Classic AUTOSAR architecture, enabling modular software development
- Improved TR software quality by increasing **MIL/SIL** test coverage using decision, condition and MC-DC coverage criteria
- Executed **HIL testing**, performing issue reproduction, log analysis and root cause analysis using ASAM MDF and Canalyzer
- Designed, developed, and integrated a **Wheel Hop Detection** algorithm in **MATLAB/Simulink** within the existing AutoTR software
 - Implemented **IMU** based tire-ground contact loss detection and benchmarked it against height-sensor-based detection model
 - Tested the model against real vehicle test data collected over different uneven terrains (speedbumps, humpback bridges, etc.)
 - Resolved a customer-facing issue causing **incorrect AutoTR mode changes** in coil-suspension variants of JLR vehicles
- Completed **Functional Safety** training with exposure to automotive safety and **ISO 26262** software development practices
- Gained industry exposure through technical visit to Global Automotive Research Centre and Motion in Control Conference

Robotics and Control Engineer | xTerra Robotics

May 2023 – May 2024

Mobile Robotics Laboratory | Advisor: Prof. Shakti Singh Gupta, Dept. of Mechanical Engineering, IIT Kanpur

- Developed an **ADAMS-MATLAB** co-simulation framework to validate planning and control algorithms for a **quadruped robot**
- Executed reference trajectories from a motion planner using a controller employing **Rigid Body Dynamics** (RBD)
- Calculated **Impact Contact Model** parameters (stiffness, damping, penetration), ensuring realistic contact behaviour for simulation
- Optimized friction and stiction parameters reducing errors in joint angles, base position and orientation, and contact forces
- Implemented **Torque-feedback-based Inverse Dynamics Controller** and **Balance Force Controller** for stable stance and trotting

Control Research Intern | SURGE Program

May 2021 – Dec 2021

Intelligence Guidance and Control Laboratory | Prof. Mangal Kothari, Dept. of Aerospace Engineering, IIT Kanpur

- Designed and tested a PZT based braking system for attitude stabilization of **dual-spin** systems under varying voltage conditions
- Simulated **Bouc-Wen model** in MATLAB/Simulink to analyze hysteresis, determine actuator stability and response time
- Implemented hybrid control using a **PID feedback controller** and a **Luenberger-observer-based feedforward controller**

KEY PROJECTS

Low-Level Controller Design for Brushless Direct Current (BLDC) Motor | Undergraduate Project Aug 2022 – Nov 2022

- Controlled position, velocity, and torque of a BLDC motor for **quasi-direct-drive** actuators widely used in quadruped robots
- Studied **Field Oriented Control** (FOC), the optimal algorithm for BLDC motor control, and worked on its implementation
- Simulated a three-phase inverter, implemented switching patterns, and open-loop FOC for speed control in Proteus

Acquisition of Modern Global Navigation Satellite System (GNSS) Signals | Course Project

Jan 2022 – May 2022

- Implemented scripts for generating **Pseudo Random Noise** (PRN) codes for GPS satellites and studied modern GNSS signals
- Employed **Fast Fourier Transform** (FFT) to analyze PRN sequence properties under noisy conditions with signal delay
- Identified satellite PRN, carrier frequency, and signal phase difference using **Parallel Code-Phase Search Acquisition** algorithm

ACHIEVEMENTS

- Secured overall **3rd position** in **Invention Factory Program** among the Top 10 selected teams – winning a prize of **INR 50,000**
- Achieved an **All-India Rank of 4419** in the JEE Advance, 2019, in the common merit list among 0.2 million shortlisted candidates
- Won **Bronze** medal in the Under-19 Chess Championship at the Council's National Sports & Games 2018 at the regional level

PATENTS

- “Squat-Sit Commode”** – Indian Design Patent No. 376349-001 – Date of Issue: July 3, 2023.
- “Manhole Cleaning Machine”** – (Pending) US Patent Application No. 63438495 – Date of Application: Jul 25, 2022.

TECHNICAL SKILLS

Programming Languages	Software	Libraries
C Python Java $\LaTeX$	MATLAB Simulink MSC ADAMS SolidWorks Proteus ROS	NumPy Pandas Matplotlib

RELEVANT COURSES

Robot Manipulators: Dynamics & Control	Vibration and Control	Basics of Modern Control Systems
Integrated Product Development	Applied Numerical Methods	Robot Motion Planning
Embedded and Cyber-Physical Systems	Machine Learning for Engineers	Autonomous Navigation